

(a) Research Interests

My research interests encompass Computer Vision, Deep Learning, Pattern Recognition, Visual Surveillance, and Multimedia Analysis, with a focus on developing interpretable, efficient, and scalable visual intelligence systems. I aim to advance foundational understanding in automated visual perception and multimodal learning, while contributing to open research resources and promoting innovation in AI systems designed for societal benefit. With over 10 years of academic and applied research experience, my work primarily addresses challenges in Person Re-identification, Human Activity Recognition, Visual Anomaly Detection, and Fine-Grained Classification. I have authored more than 15 peer-reviewed publications in internationally reputed journals and conferences, including IJCV, IEEE TAI, Neurocomputing, and ICIP. My research at both postdoctoral positions has been supported by highly competitive EPSRC (UKRI) research grants in the UK, focusing on real-time vision systems for autonomous and intelligent systems and smart healthcare applications. I also serve as an active reviewer for leading international journals and conferences, reflecting continued recognition and engagement within the global computer vision research community.

(b) Research Experience

1. Academic

Year	Research Activity
2023 – 2026	Postdoctoral Research Associate in EPSRC funded project “ATTRACT - A Trustworthy Robotic Autonomous system to support”, Edge Hill University, UK. PI: Prof. Ardhendu Behera. <i>Leading the design and development of a drone-based autonomous triage system for high-risk environments using multimodal AI, real-time vital sign estimation, and injury severity prediction. Focused on integrating trustworthy AI principles to ensure ethical, legal, and operational reliability under MoD and WHO-aligned frameworks.</i>
2022 – 2023	Postdoctoral Research Associate in EPSRC funded project SPHERE-A (Sensor Platform for HEalthcare in a Residential Environment (IRC Next Steps)), University of Bristol, UK. PI: Prof. Majid Mirmehdi. <i>Led the vision-based research component of a smart home sensing platform for chronic health monitoring. Developed a deep learning pipeline for estimating turning angles from monocular in-home videos using 3D pose estimation, enabling remote and unobtrusive symptom assessment in Parkinson’s disease.</i>

<i>Year</i>	<i>Research Activity</i>
2015 – 2021	PhD in Computer Vision-Based Video Surveillance, Jadavpur University, Kolkata, India. Supervisor: Prof. Ananda S. Chowdhury. Thesis: “On Some Problems in Computer Vision Based Video Surveillance” <i>Thesis addressed challenges in real-time anomaly detection, crowd behavior analysis, and person re-identification by proposing lightweight, training-less methods suited for resource-constrained surveillance environments.</i>
2013 – 2015	M.Tech at Indian Institute of Technology, Bhubaneswar, Orissa, India. Supervisor: Dr. Debi Prosad Dogra. Thesis: Computer Vision-Based Pulse Rate Detection <i>Thesis focused on developing a contactless vision-based method to estimate pulse rate in infants during neurological exams, leveraging color signal analysis and dimensionality reduction to enhance comfort and reliability over traditional sensors.</i>
2. Professional	
2020 – 2020	Project on “UAV Guided Helicopter Landing” in collaboration with ELMAX Systems and Solutions, supervised by Prof. Ananda S. Chowdhury. <i>Built a vision-guided navigation framework to autonomously identify safe landing zones for helicopters in dynamic environments.</i>
2021 – 2021	“Artificial Intelligence / Machine Learning Engineer”, National Informatics Centre (NIC), Ministry of Electronics and Information Technology (MeitY), India. <i>Developed a face-recognition-based (FRS) attendance monitoring system for the Utkarsh Bang initiative and contributed to AI frameworks for road inspection, traffic congestion analysis, and helmetless two-wheeler detection using deep learning and computer vision.</i>

(c) Education

Ph.D. (2021)	Electronics and Telecommunication Engineering, Jadavpur University, Kolkata, India <i>CGPA: 9.5/10</i>
M.Tech (2015)	Electronics and Communication Engineering, Indian Institute of Technology Bhubaneswar, Odisha, India <i>CGPA: 8.16/10, Division: First Class</i>
B.E. (2013)	Electronics and Telecommunication Engineering, Jadavpur University, Kolkata, India <i>CGPA: 8.06/10, Division: First Class</i>
Higher Secondary (2009)	Science (Physics, Chemistry, Mathematics, Computer Science), Bidhan Chandra Institution, Durgapur

Secondary (2007) **Marks: 78%, Division: First Class**
General (English, Bengali, Mathematics, Science, Social Science, IT), D.A.V. Model School, Durgapur
Marks: 86%, Division: First Class

(d) List of Publications

(Book Chapter)

- B1. **A. Sikdar**, A. S. Chowdhury, “*Efficient Deep Learning for Key Challenges in Person Re-identification*”, in *Pattern Recognition and Computer Vision in the New AI Era*, World Scientific, 2025, pp. 399–425.

(International Journals)

- J1. A. Humnabadkar, **A. Sikdar**, B. Cave, H. Zhang, N. Bessis, A. Behera, “*From Virtual Environments to Real-World Trials: Emerging Trends in Autonomous Driving*”, *IEEE Trans. Intell. Transp. Syst.*, 27(8), 8901–8921, 2026
- J2. Q. Cheng, C. Morgan, **A. Sikdar**, A. Masullo, A. Whone, and M. Mirmehdi, “*Your turn: At home turning angle estimation for Parkinson’s disease severity assessment*”, *Artificial Intelligence in Medicine* p.103194.
- J3. **A. Sikdar**, Y. Liu, S. Kedarisetty, Y. Zhao, A. Ahmed and A. Behera, “*Interweaving Insights: High-Order Feature Interaction for Fine-Grained Visual Recognition*”, *Int. J. Comput. Vis.*, 133 (4), 1755-1779.
- J4. **A. Sikdar** and A. S. Chowdhury, “*Lightweight Learning for Partial and Occluded Person Re-identification*”, *IEEE Trans. Artif. Intell.*, vol. 5, no. 6, pp. 3245-3256, June 2024
- J5. C. Morgan, E. L. Tonkin, A. Masullo, F. Jovan, **A. Sikdar**, P. Khaire, M. Mirmehdi *et. al.* “*A multimodal dataset of real world mobility activities in Parkinson’s disease*”, *Scientific Data*, 10(1), 918.
- J6. **A. Sikdar** and A. S. Chowdhury: “*An Adaptive Training-less Framework for Anomaly Detection in Crowd Scenes*”, *Neurocomputing* 415 (2020), 317 - 331.
- J7. **A. Sikdar** and A. S. Chowdhury: “*Scale-Invariant Batch-Adaptive Residual Learning for Person Re-identification*”, *Pattern Recognit. Lett.* 129 (2020), 279 - 286.
- J8. **A. Sikdar**, S. K. Behera, and D. P. Dogra: “*Computer-vision-guided human pulse rate estimation: a review.*” *IEEE Rev. Biomed. Eng.* 9 (2016): 91-105.

(International Conferences and Workshops)

- C1. A. Humnabadkar, **A. Sikdar**, H. Zhang, T. Hussain, A. Behera: “*Driving Through Graphs: A Bipartite Graph for Traffic Scene Analysis*”, *IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, UAE (2024), pp. 908–914.
- C2. A. Humnabadkar, **A. Sikdar**, B. Cave, H. Zhang, P. Bakaki, A. Behera: “*High-Order Evolving Graphs for Enhanced Representation of Traffic Dynamics*”, *European Conference on Computer Vision (ECCV) – 2nd Workshop on Vision-Centric Autonomous Driving (VCAD)*, Milan, Italy (2024).
- C3. **A. Sikdar**, A. Sahu, D. Sen, R. Mahajan, A.S. Chowdhury: “*Unconstrained Vision Guided UAV Based Safe Helicopter Landing*”, *International Conference on Pattern Recognition (ICPR)*,

Milan, Italy (2021), pp. 8328-8335.

- C4. **A. Sikdar**, D. Chatterjee, A. Bhowmik, A. S. Chowdhury: “*Open-Set Metric Learning for Person Re-Identification in the Wild*”, IEEE International Conference on Image Processing (ICIP), Abu Dhabi, UAE (2020), pp. 2356-2360.
- C5. **A. Sikdar**, A. S. Chowdhury: “*Multi-Level Threat Analysis in Anomalous Crowd Videos*”, International Conference on Computer Vision and Image Processing (CVIP), N. Nain et al. (eds.), Springer Communications in Computer and Information Science, Vol. 1148, Jaipur, India (2019), pp. 495 - 506.
- C6. **A. Sikdar**, A. S. Chowdhury: “*An Ellipse Fitted Training-less Model for Pedestrian Detection*”, IEEE International Conference on Advances in Pattern Recognition (ICAPR), Bangalore, India (2017), pp. 1-6.
- C7. R. Roy, **A. Sikdar**, S. Mazumder, A. S. Chowdhury: “*ESAC: An Algorithm for Fissure Line Detection*”, IEEE International Conference on Advances in Pattern Recognition (ICAPR), Bangalore, India (2017), pp. 1-6.
- C8. **A. Sikdar**, S. K. Behera, D. P. Dogra, and H. Bhaskar. “*Contactless vision-based pulse rate detection of Infants Under Neurological Examinations.*”, IEEE International Conference on Engineering in Medicine and Biology Society (EMBC), Milan, Italy (2015), pp. 650-653.

(e) Other Professional Experiences

- 1. Selected for the highly competitive *International Computer Vision Summer School (ICVSS 2025)* held in Sicily, Italy, and awarded Certificates of Attendance and Exam Qualification.
- 2. Selected to present and participate in the “*Human and Computer Models Workshop*” held at the University of Surrey, UK, on May 15, 2024.
- 3. Presented the paper “*Open-Set Metric Learning for Person Re-Identification in the Wild*” at the 27th *IEEE International Conference on Image Processing (ICIP 2020)*, Abu Dhabi, UAE, October 25–28, 2020.
- 4. Presented the paper “*Multi-Level Threat Analysis in Anomalous Crowd Videos*” at the 4th *International Conference on Computer Vision and Image Processing (CVIP 2019)*, MNIT Jaipur, India, September 27–29, 2019.
- 5. Presented the paper “*An Ellipse Fitted Training-less Model for Pedestrian Detection*” at the 9th *International Conference on Advances in Pattern Recognition (ICAPR 2017)*, Indian Statistical Institute (ISI) Bangalore, India, December 28–30, 2017.
- 6. Participated in the “*Workshop for Presentation of Research Work of Visvesvaraya PhD Scholars*”, conducted by the Ministry of Electronics and Information Technology (MeitY), MNIT Jaipur, India, September 13–15, 2018.
- 7. Participated in the *Third Summer School on Computer Vision*, conducted by IIIT Hyderabad, India, July 2–7, 2018.

(f) Technical Skills

- 1. Proficient in *MATLAB* and *Python programming*, with extensive experience in implementing algorithms for computer vision, signal processing, and deep learning research applications.
- 2. Hands-on expertise in leading deep learning and scientific computing frameworks, including *TensorFlow*, *PyTorch*, *Keras*, *NumPy*, *SciPy*, and *Scikit-learn*, for model development, optimization, and experimental analysis.
- 3. Skilled in end-to-end AI research workflows, encompassing *data preprocessing*, *model de-*

sign, evaluation, and reproducibility for large-scale visual recognition and pattern analysis tasks.

4. Basic working knowledge of *Python-based GUI development using PyQt5*, applied to developing interactive tools for visualization and user-driven experimentation.

(g) Professional Activities

1. *Led research contributions to the EPSRC-funded project “ATRACT (A Trustworthy Robotic Autonomous System to Support Casualty Triage)”* at Edge Hill University, UK, engaging in interdisciplinary research discussions and supporting the development of multi-modal AI systems for autonomous triage and decision support.
2. *Design and led the development of lecture content and seminar activities* for the Emerging Technologies module (CIS4518, Level 7) at Edge Hill University. Designed and delivered Google Colab-based interactive notebooks demonstrating the use of NumPy, Matplotlib, and TensorFlow to build and run deep learning frameworks, thereby enhancing the practical learning experience of postgraduate students.
3. *Coordinated cross-disciplinary collaboration within a research team at the University of Bristol*, bridging clinical experts and computer scientists to systematically integrate domain-specific feedback into structure-aware computer vision models. This effort culminated in the development of robust visual understanding frameworks and a peer-reviewed publication.
4. Active reviewer for leading international journals including *IEEE Transactions on Image Processing (TIP)*, *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, *Computer Vision and Image Understanding (CVIU)*, *Pattern Recognition Letters (PRL)*, and *Signal Processing Letters (SPL)*, as well as premier conferences venues such as *CVPR*, *BMVC*, *CVIP*, and *ICAPR*.

(h) Teaching and Mentoring

1. Developed and delivered lectures on “*Introduction to Computer Vision*” as part of the *Emerging Technologies module* (CIS4518, Level 7) at Edge Hill University, UK, offered to MSc Computing, MSc Artificial Intelligence, and LLM students. The lectures covered key topics in digital image representation, filtering, convolution operations, and AI ethics, combining theoretical concepts with practical demonstrations in a research-led teaching framework.
2. Led lecture delivery and served as substitute lecturer for the Advanced Robotics and Artificial Intelligence module (CIS3137, Level 6) at Edge Hill University, independently developing and delivering sessions on computer vision, image processing, and deep learning for autonomous systems. Drove both curriculum development and module delivery, aligning learning outcomes with cutting-edge research and industry practice.
3. Served as Guest Faculty at Jadavpur University, teaching undergraduate laboratory courses in Digital Signal Processing Lab, Data Structures Lab, and Microprocessor & Microcontroller Lab (B.E. Electronics and Telecommunication Engineering). Designed and supervised structured laboratory exercises that strengthened students’ understanding of algorithms, signal processing, and hardware interfacing.
4. Actively involved in supervision and mentorship at undergraduate and postgraduate levels, including M.Res and Ph.D. students. Provided research guidance and mentorship to students and research assistants working on AI-related projects, leading to publications, conference presentations, and project-based outcomes in computer vision and deep learning domains.